MCTGCL: Mixed CNN–Transformer for Mars Hyperspectral Image Classification With Graph Contrastive Learning

Bobo Xi, *Member, IEEE*, Yun Zhang, *Graduate Student Member, IEEE*, Jiaojiao Li, *Senior Member, IEEE*, Tie Zheng, Xunfeng Zhao, Haitao Xu, Changbin Xue, Yunsong Li, *Member, IEEE*, and Jocelyn Chanussot, *Fellow, IEEE*

***Abstract*— Hyperspectral image (HSI) classification has been extensively studied in the context of Earth observation. However, its application in Mars exploration remains limited. Although convolutional neural networks (CNNs) have proven effective in HSI processing, their local receptive fields hinder their ability to capture long-range features. Transformers excel in global modeling and perform well in HSI classification (HSIC), but they often neglect the effective representation of local spectral and spatial features and tend to be more complex. To address these challenges, we propose a mixed CNN–transformer network for Mars HSI classification with graph contrastive learning to enhance classification performance. Specifically, we introduce an information-enhanced attention module (IEAM) designed to aggregate attention features from multiple perspectives. Additionally, we develop a lightweight dual-branch CNN–transformer (LDCT) network that efficiently extracts both local and global spectral–spatial features with lower complexity. To improve the discrimination of inter-class features, we apply graph contrastive learning to the topological structure of labeled samples. Furthermore, we annotated three Mars HSI datasets, referred to as HyMars, to validate the effectiveness of our proposed mixed CNN–transformer network for Mars HSIC with graph contrastive learning (MCTGCL). Comprehensive experimental results across different amounts of labeled samples consistently demonstrate the superiority of the method. The source code is available at https://github.com/B-Xi/TGRS_2025_MCTGCL.**

***Index Terms*— Graph contrastive learning, information-enhanced attention, lightweight, Mars exploration.**

I. Introduction

CLASSIFICATION of surface minerals on Mars is essential for comprehending changes in the Martian atmospheric and geological conditions [1], [2]. By identifying and analyzing these minerals, we can gain valuable insights into the environmental state of Mars, its geological evolution, and its potential for future habitability. Hyperspectral image (HSI) captures extensive spectral information across hundreds of continuous channels, allowing the differentiation of various ground objects based on their spectral signatures [3], [4], [5]. This high spectral resolution enables the mapping of mineral distributions and regions on the Martian surface [6].

Compared to commonly studied remote sensing HSIs used for Earth observation, the classification of Mars HSIs presents unique and unresolved challenges. The primary focus for Mars HSI analysis is the identification and classification of mineral types, which are characterized by high spectral similarities, making them difficult to differentiate. Additionally, the task is further complicated by the limited availability of representative samples for training, as well as the variations in spectral signatures caused by factors such as atmospheric interference, lighting conditions, and surface composition. These challenges underscore the need for advanced algorithms and methodologies tailored to the specific complexities of Martian hyperspectral data.

Traditional HSI classification (HSIC) methods have primarily focused on the spectral characteristics of the imagery, utilizing techniques such as support vector machines (SVMs) [7] and extreme learning machines (ELMs) [8]. Additionally, dimensionality reduction methods such as principal component analysis (PCA) [9] and linear discriminant analysis (LDA) [10] have been employed. However, these conventional approaches often struggle to capture subtle image details, leading to lower classification accuracy. Furthermore, many traditional feature extraction methods depend on specialized knowledge for manual feature crafting, which tends to be shallow. This

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Bobo Xi, Jiaojiao Li, and Xunfeng Zhao are with the State Key Laboratory of Integrated Service Networks, School of Telecommunications Engineering, Xidian University, Xi'an 710071, China, and also with the National Space Science Center, Chinese Academy of Sciences, Beijing 100190, China (e-mail: xibobo@xidian.edu.cn; jjli@xidian.edu.cn; zhaoxf@nssc.ac.cn).

Yun Zhang and Yunsong Li are with the State Key Laboratory of Integrated Service Networks, School of Telecommunications Engineering, Xidian University, Xi'an 710071, China (e-mail: zzzz358@foxmail.com; ysli@mail.xidian.edu.cn).

Tie Zheng, Haitao Xu, and Changbin Xue are with the Key Laboratory of Electronics and Information Technology for Space Systems, National Space Science Center, Chinese Academy of Sciences, Beijing 100190, China (e-mail: zhengtie@nssc.ac.cn; xuhaitao@nssc.ac.cn; xuechangbin@nssc.ac.cn).

Jocelyn Chanussot is with the Université Grenoble Alpes, Inria, CNRS, Grenoble INP, LJK, 38000 Grenoble, France (e-mail: jocelyn@hi.is).

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reliance can hinder the model's generalization performance and its ability to adapt to new complex tasks [11].

In recent decades, artificial neural networks have demonstrated an impressive ability to continuously learn and extract significant features from raw data, resulting in a growing emphasis on HSIC utilizing deep learning techniques [12], [13], [14], even in the more complex open-set conditions [15], [16], [17], [18]. Particularly, a variety of network architectures have been employed for HSIC, including convolutional neural networks (CNNs), deep belief networks (DBNs), recurrent neural networks (RNNs), and graph convolutional networks (GCNs), among others. Notably, CNNs have become particularly popular due to their fundamental properties of local connectivity and weight sharing, which enable them to effectively extract rich spectral and spatial features from hyperspectral data. For instance, Yu et al. [19] introduced a straightforward yet effective 2-D CNN model designed to extract valuable features for precise HSIC. Similarly, Li et al. [20] developed a 3-D CNN that processes HSIs in both spatial and spectral dimensions, achieving improved classification outcomes. Following this, Zhong et al. [21] created an end-to-end spectral–spatial residual network (SSRN) that captures discriminative features of the extensive spectral characteristics and spatial context. With the rise of various attention-based techniques, such as squeeze-and-excitation (SE) attention [22] and efficient channel attention (ECA) [23] in computer vision, there has been increasing interest in applying attention mechanisms to HSIC. For example, an attention-based adaptive spectral–spatial kernel-improved residual network (A^2S^2K-ResNet) was proposed to effectively capture discriminative spectral–spatial features through end-to-end training [24]. However, CNNs face limitations due to their local receptive fields and complex convolutions, which restrict their capacity to capture global spatial information in HSIs and effectively highlight different spectral–spatial features [25].

More recently, transformer-based classification methods have gained considerable attention. Unlike CNNs, transformers excel at capturing long-term dependencies through multi-head self-attention (MHSA), a key component that prioritizes significant features while diminishing the less important ones. For HSIC, several innovative transformer-based approaches have been introduced. For instance, Hong et al. [26] designed a novel transformer backbone network called SpectralFormer, which integrates group-wise spectral embedding and cross-layer adaptive fusion modules to learn detailed local spectral representations and convey memory-like components from shallow to deep layers. Additionally, a group-aware hierarchical transformer (GAHT) [27] was proposed, which restricts MHSA to the local spatial–spectral context by incorporating a new grouped pixel embedding module. Furthermore, Xue et al. [28] developed a local transformer with a spatial partition restore network (SPRLT-Net), treating each pixel as a token to thoroughly and accurately extract detailed spatial relationships, particularly for boundary pixels.

Despite the impressive performance of various transformer-based methods, they still face challenges in effectively extracting local features such as lines, corners, and edges within images. This limitation hinders further advancements in HSIC performance. To address this issue, Yang et al. [29] introduced an HSI transformer (HiT) classification network that incorporates convolution operations into the transformer, enabling it to capture subtle spectral variations and convey local spatial context. Additionally, Zhang et al. [30] proposed a local–global cross-fusion network that integrates 3-D CNNs, transformer feature extraction, and feature fusion, successfully capturing both local and global spectral–spatial information from HSIs. Similarly, Zhang et al. [31] developed a convolutional transformer mixer (CTMixer) that combines CNNs and transformer architectures to extract both global and local hyperspectral features. Furthermore, Jiang et al. [32] introduced a graph generative structure-aware transformer (GraphGST) that aggregates local-to-global correlations by incorporating graph structural information into the transformer design.

Inspired by these valuable studies for Earth observation, we focus on the fusion of CNN and transformer networks. A key challenge is the large number of parameters and computational demands of the model, making the development of a lightweight and efficient CNN–transformer module essential. Additionally, when comparing SE attention, which explicitly models cross-dimensional interactions for channel-wise attention extraction [22], coordinate attention (CA) [33] can embed direction-specific information into channel attention along the spatial dimension, leading to improved feature aggregation and competitive performance in computer vision. We focus on designing a relatively lightweight module based on this attention mechanism for HSIC.

Lately, contrastive learning has emerged as a prominent research area in image classification tasks. The objective is to train networks to effectively learn a feature space where similar samples are closely mapped together, while dissimilar samples are positioned further apart. For instance, Xue et al. [34] developed a weighted contrastive loss to guide the model in the right direction by emphasizing sample pairs that are difficult to distinguish. Although contrastive learning has proven effective for HSIC, challenges remain. Many previous studies have relied on traditional convolutional structures for extracting information from HSIs, often overlooking the inherent graph-structured information of the samples.

Building on the previous analysis, we propose a mixed CNN–transformer network for Mars HSIC with graph contrastive learning (MCTGCL). This approach leverages the advantages of CNNs and transformers while maintaining a lightweight design. To begin with, we develop an information-enhanced attention module (IEAM) that integrates spatial positional information into channel attention maps, thereby improving feature aggregation. This is achieved by applying average pooling (AP) across different axes and merging attention vectors from two branches. Next, we present a lightweight dual-branch CNN–transformer (LDCT) that extracts local and global features simultaneously, reducing the overall complexity of the model. Furthermore, to enhance the embedding of discriminative features and emphasize the inherent graph-structured information of HSI, we input the

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feature embeddings into GCNs and create a supervised graph contrastive loss function to facilitate practical network training.

The main contributions of our proposed framework can be summarized as follows.

1) We design the IEAM to consolidate attention information from various aspects. To preserve information within each channel and minimize computational demands, we reshape partial channels into batch dimensions and organize the channel dimensions into multiple sub-features.
2) Rather than applying CNN and transformer to the entire patch, we create an LDCT that divides the input along the channel dimension. One part undergoes CNN processing, while the significant information of this part and the complete features from the other part are processed by the transformer.
3) To accurately capture the topological structure of samples, we introduce a supervised graph contrastive loss that reduces the distance between prototypes of the same category across different views while increasing the separation between prototypes of different categories.
4) We establish three benchmark Martian HSI datasets (HyMars), and perform experiments on them. The classification results across various training samples consistently demonstrate the effectiveness of our proposed frameworks.

The remainder of the article is organized as follows. Section II provides a brief overview of related works. Section III details our proposed method. Section IV presents extensive experiments and analyzes the classification results. Section V draws the conclusions.

II. Related Works

A. Mars HSIC

The identification and mapping of minerals on Mars offer crucial insights into the planet's historical climate and geological evolution, as well as the processes that have shaped its surface [35]. Consequently, HSIC for Mars has garnered increasing attention in recent research.

Initially, techniques based on distance-based similarity matching, such as Euclidean distance (ED) [36] and spectral angle mapper (SAM) [37], were utilized for mineral mapping due to their simplicity and efficiency in classifying large volumes of hyperspectral data. However, these methods often fell short of capturing semantic differences, making them less effective for accurate mineral identification. As a result, alternative methods, including probabilistic, unsupervised, and neural network approaches, have been developed to overcome these limitations and show enhanced effectiveness. Neural networks, in particular, are well-known for their capability to reveal hidden patterns within data. For example, Caggiano et al. [38] applied CNNs to classify specific mineral assemblages on the Martian surface. Additionally, Saranathan and Parente [39] utilized generative models to enhance the distinguishability of spectral data, demonstrating that features obtained through generative adversarial networks (GANs) were more effective in differentiating mineral signatures compared to traditional summary parameters and similarity metrics.

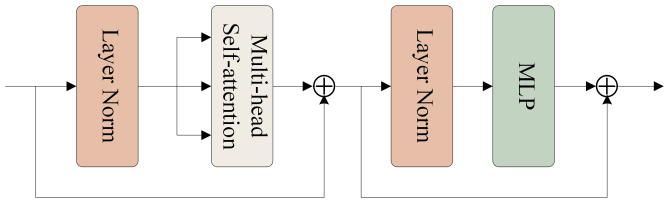


Fig. 1. Architecture of transformer encoder block.

Nonetheless, these deep learning methods typically require a significant amount of labeled training data, which poses a challenge for mineral identification on Mars due to the limited availability of ground-truth and training datasets. This scarcity hampers the development of supervised learning models for this application. Currently, only a handful of studies have employed neural network-based techniques for mineral exploration. To address the challenge of limited labeled samples, we have created three datasets representing typical Martian areas for future research. Furthermore, we have designed a neural network specifically for Mars HSIC and validated its effectiveness using these datasets under small sample sizes.

B. Transformer Mechanism for HSIC

In the past few years, the success of transformers in natural language processing (NLP) has significantly influenced computer vision and has spurred the development of HSIC [40], [41]. As illustrated in Fig. 1, transformer encoder blocks utilizing MHSA are employed to model global dependencies across long-range positions and bands within an HSI patch.

The initial research in HSIC, as noted in [42], involved the direct application of bidirectional encoder representations from transformers (BERTs) on flattened input patches. Following this, a variety of transformer-based approaches have emerged. For example, to enhance the representation of high-level semantic features and capture spectral–spatial characteristics, Sun et al. [43] integrated CNNs with transformers to propose a spectral–spatial feature tokenization transformer (SSFTT), which utilizes a Gaussian-weighted feature tokenizer for extracting higher-level sequence features. To effectively extract deeper spatial–spectral and cross-channel features, Qiu et al. [44] introduced a cross-channel dynamic spatial–spectral fusion transformer (CDSFT). Additionally, Roy et al. [45] presented a novel morphological transformer (morphFormer) that incorporates a learnable spectral and spatial morphological network, employing spectral and spatial morphological convolution operations alongside an attention mechanism to enhance the interaction between the structural and shape information of HSI tokens and classification tokens. Shi et al. [46] proposed a dual-branch multiscale transformer (DBMST) that leverages the strengths of transformers and multiscale features, utilizing two branches for feature extraction at different scales to achieve richer discriminative features.

This article proposes a mixed CNN–transformer network for Mars HSIC. Different from conventional attention-based HSIC methods that primarily focus on the channel dimension,

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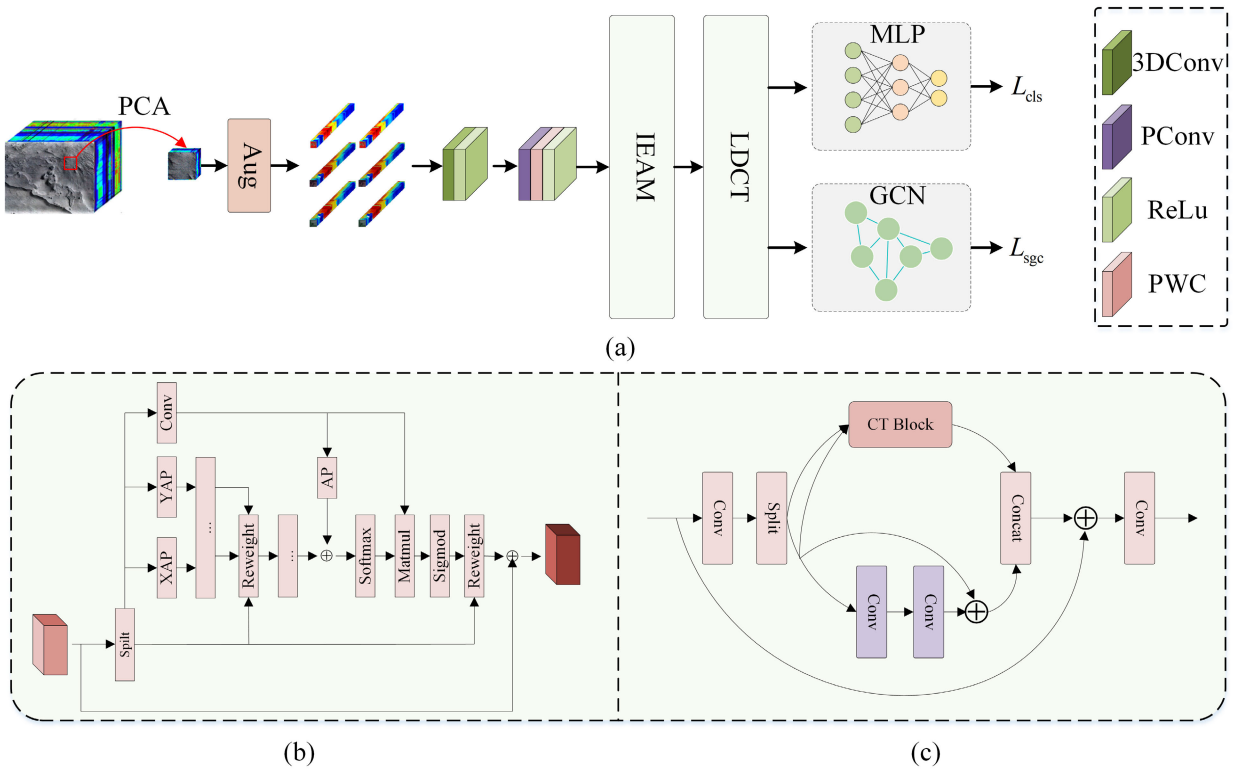


Fig. 2. Detailed frameworks of the proposed MCTGCL. (a) Architecture of the MCTGCL comprises IEAM and LDCT. Specifically, a hybrid loss, including classification loss L_{cls} and supervised graph contrastive loss L_{sgc}, is presented to train the network. (b) Structure of the IEAM. (c) Structure of the LDCT.

our approach leverages IEAM to integrate spatial positional information into channel attention maps and generate features with diverse information. Besides, we design a light dual-branch CNN–transformer termed LDCT. Instead of applying the CNN and transformer to the whole patch or designing a serial CNN–transformer module, the LDCT divides the data into two sub-maps and extracts local and non-local features concurrently, which is relatively lightweight and effective.

C. Contrastive Learning for HSIC

The fundamental concept of contrastive learning is to minimize the distance between similar samples while maximizing the distance between samples from different categories. Typically, contrastive learning employs various loss functions to define the distance between similar and dissimilar samples, including supervised, unsupervised, and semi-supervised contrastive losses.

Numerous studies have explored contrastive learning in the context of HSIC. For instance, an adversarial domain alignment method combined with contrastive learning has been developed to ensure feature consistency, facilitating knowledge transfer [47]. To capture essential representations of both source and target data, contrastive learning is applied in a self-supervised manner to enhance robustness and sensitivity, utilizing a wealth of unlabeled samples through data augmentation, where augmented data from the same or different samples is encouraged to be close or distant, respectively. In [48], to improve instance discriminability, the spectral–spatial masked transformer (SS-MTr) was integrated with contrastive learning, resulting in a new HSIC approach known as contrastive SS-MTr (C-SS-MTr). This method generates plausible contrastive views using masked and unmasked HSI inputs. By maintaining locality in modeling, C-SS-MTr effectively learns instance discrimination by differentiating each HSI training sample from the others in the contrastive learning framework.

III. Proposed Method

The architecture of the proposed method is depicted in Fig. 2. Initially, the PCA is applied to the original HSI to effectively reduce its dimensionality. The resulting lower-dimensional data is then divided into small 3-D patches, with each patch augmented into two views, which serve as inputs for the feature extraction module. Following this, a shallow and lightweight CNN module is utilized to extract combined spectral and spatial features. We then introduce the IEAM to embed spatial information into channel attention, followed by an LDCT that processes partial data in parallel. Finally, we implement a hybrid loss function that includes classification loss L_{cls} and supervised graph contrastive loss L_{sgc} to train the network.

A. Shallow CNN Module

The CNNs have been widely used in HSIC tasks due to their remarkable ability to extract local features. In our proposed network, we integrate a shallow CNN designed to effectively capture spectral and spatial information from HSI. This module consists of a 3-D convolutional (3DConv) block, followed

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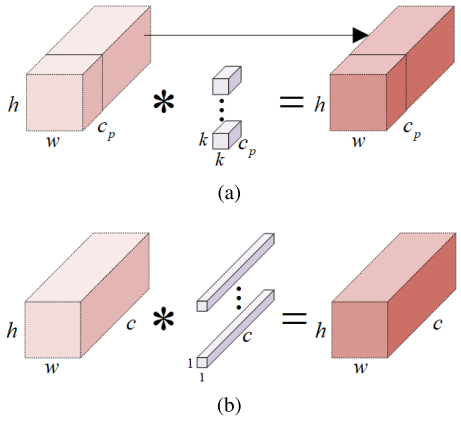


Fig. 3. (a) Partial convolution. (b) Point-wise convolution.

by a partial convolution (PConv) and a point-wise convolution (PWC) in sequence. The 3DConv block is aimed explicitly at extracting initial spatial–spectral information, while PConv and PWC are utilized to further exploit spatial and spectral information, respectively.

As shown in Fig. 3(a), PConv is a lightweight and efficient operation that minimizes frequent memory access, which is a primary contributor to low floating-point operations per second (FLOPs). It leverages redundancy within the feature maps by applying standard convolutions to only a subset of input channels, leaving the others untouched. This approach reduces both redundant computations and memory access. Assuming that regular convolution filters with a kernel size of $k \times k$ are applied to the input $\mathbf{X}_I \in \mathbb{R}^{h\times w\times c}$ to produce the output $\mathbf{X}_O \in \mathbb{R}^{h\times w\times c}$, the FLOPs for this operation and the number of memory access can be calculated as $h \times w \times k^2 \times c^2$ and $h \times w \times 2c + k^2 \times c^2$, respectively. In contrast, PConv has lower FLOPs, calculated as $h \times w \times k^2 \times {c_p}^2$, and requires less memory access, quantified as $h \times w \times 2c_p + k^2 \times {c_p}^2$, compared to regular convolution, where c_p denotes the number of channels being processed. To fully and efficiently utilize information from all channels, we further incorporate a PWC following the PConv. The architecture of the PWC is illustrated in Fig. 3(b), facilitating the flow of feature information across all channels.

B. Information-Enhanced Attention Module

Unlike the traditional SE module that utilizes global AP to model cross-channel information, we introduce an IEAM to merge attention maps through two parallel subnetworks, as shown in Fig. 4. Initially, we segment the input into G sub-features along the channel dimension to capture diverse semantics while reducing computational load. These groups are represented as $\mathbf{X} = [\mathbf{X}_0, \mathbf{X}_1, \ldots, \mathbf{X}_{G-1}]$, where $\mathbf{X}_g \in \mathbb{R}^{H\times W\times (C/G)}$. In the left 1×1 branch, we implement two 1-D global AP operations to encode channel information along two spatial directions, represented as follows:

$$\mathbf{Z}^H = \sum_{0\le j\le W} \frac{1}{W}\mathbf{X}_g(H, j) \tag{1}$$

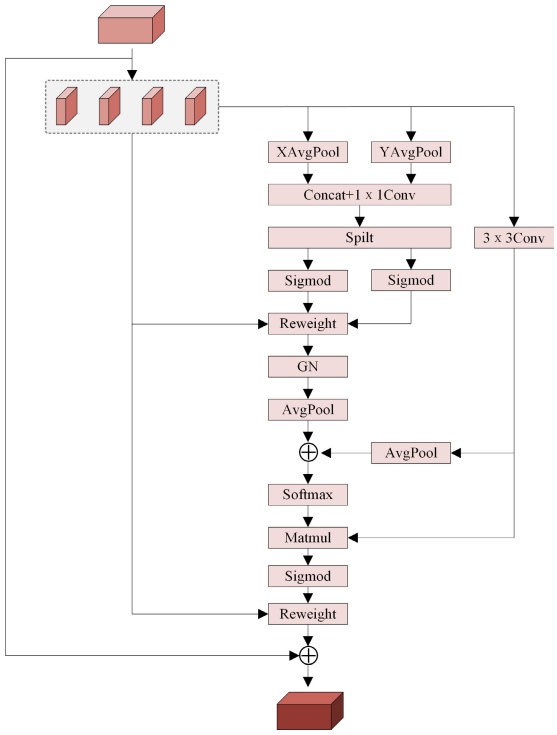


Fig. 4. Detailed architecture of the proposed IEAM.

$$\mathbf{Z}^W = \sum_{0\le i\le H} \frac{1}{H}\mathbf{X}_g(i, W) \tag{2}$$

where $\mathbf{Z}^H \in \mathbb{R}^{H\times (C/G)}$ and $\mathbf{Z}^W \in \mathbb{R}^{W\times (C/G)}$ focus on the information from different directions. $\mathbf{Z}^H$ is derived through 1-D global AP along the horizontal dimension, effectively capturing positional information across the vertical dimension and preserving precise positional information at the horizontal dimension direction. Similarly, $\mathbf{Z}^W$ is obtained through 1-D global AP along the vertical dimension, thereby encoding positional information along the horizontal dimension. Then, we concatenate the two tensors and capture local cross-channel interaction with 1×1 convolution. After decomposing the outputs of the 1×1 convolution into two vectors, we apply two non-linear sigmoid functions. Subsequently, we combine the channel-wise attention maps within each group through simple multiplication to create distinct cross-channel interactive features between the two parallel paths in the 1×1 branch.

Next, we perform a 2-D global AP in the 1×1 branch to capture global spatial information

$$\mathbf{Z}^{\mathrm{HW}} = \sum_{0\le i\le H}\sum_{0\le i\le W} \frac{1}{H \times W}\mathbf{Z}'(i, j) \tag{3}$$

where $\mathbf{Z}'$ denotes the matrix prior to 2-D AP. In this branch, we model the cross-channel dependencies with spatial positional information. Additionally, a single 3×3 kernel is utilized in the right 3×3 branch to capture features at different scales and achieve the local cross-channel interaction, along with another 2-D global AP operation. To provide better attention for high-level feature maps, we merge attention information from both branches. This is followed by matrix multiplication between the outputs of the fusion process and the results of

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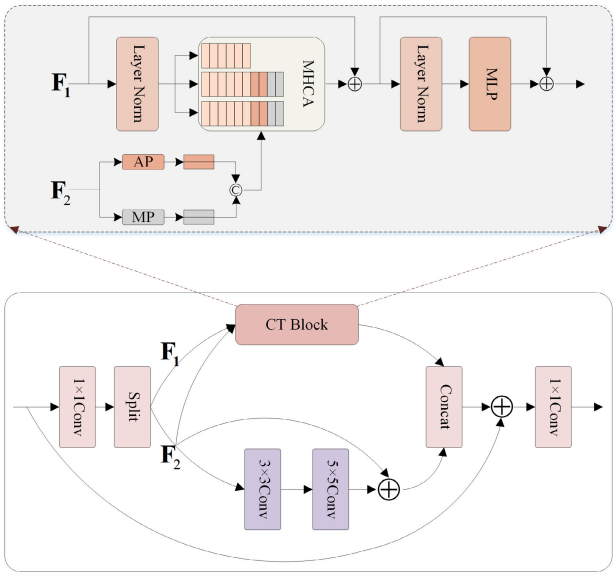


Fig. 5. Overall architecture of LDCT and the contained CT block.

the 3×3 convolution, yielding the spatial attention map after a sigmoid calculation. Finally, we multiply this attention map with $\mathbf{X}$ and introduce a shortcut to mitigate overfitting.

C. Lightweight Dual-Branch CNN–Transformer

To leverage the strengths of both CNNs and transformers while maintaining low complexity, we propose an LDCT network. This architecture combines the local modeling capabilities of CNNs with the non-local modeling abilities of transformers, as illustrated in Fig. 5.

The input $\mathbf{F}$, sized $H^F \times W^F \times C$, is first processed through a 1×1 convolutional layer producing C output channels. We then split the resulting matrix into two sub-matrices, $\mathbf{F}_1$ and $\mathbf{F}_2$, each sized $H^F \times W^F \times (C/2)$. Both $\mathbf{F}_1$ and $\mathbf{F}_2$ are fed into a novel cross transformer (CT) block, with $\mathbf{F}_2$ also directed to the CNN branch. Specifically, AP and max pooling (MP) are applied to $\mathbf{F}_2$ to retain valuable information. The AP preserves overall patterns and structural information, while the MP highlights salient feature points, suppressing less important features. These features are then utilized in a multihead cross-attention (MHCA) mechanism alongside $\mathbf{F}_1$ to generate $\mathbf{K}$ and $\mathbf{V}$. Linear transformations yield four matrices: query $\mathbf{Q} \in \mathbf{R}^{s\times d}$, key $\mathbf{K}' \in \mathbf{R}^{s\times d}$, value $\mathbf{V}' \in \mathbf{R}^{s\times d}$, and information $\mathbf{I}' \in \mathbf{R}^{l\times d}$, defined as follows:

$$\mathbf{Q} = \mathbf{F}_1' \mathbf{W}_q, \mathbf{K}' = \mathbf{F}_1' \mathbf{W}_k, \mathbf{V}' = \mathbf{F}_1' \mathbf{W}_v, \mathbf{I}' = \mathbf{F}_2' \mathbf{W}_i \quad (4)$$

where $\mathbf{F}_1'$ and $\mathbf{F}_2'$ are the inputs of the MHCA. $\mathbf{W}_q \in \mathbb{R}^{d\times d}$, $\mathbf{W}_k \in \mathbb{R}^{d\times d}$, $\mathbf{W}_q \in \mathbb{R}^{d\times d}$, and $\mathbf{W}_i \in \mathbb{R}^{d\times d}$ are trainable matrices. The matrices $\mathbf{Q}$, $\mathbf{K}'$, $\mathbf{V}'$, and $\mathbf{I}'$ are then divided into h parts along the d dimension

$$\mathbf{Q} = (\mathbf{Q}_1, \mathbf{Q}_2, \ldots, \mathbf{Q}_h) \quad (5)$$

$$\mathbf{K}' = (\mathbf{K}_1', \mathbf{K}_2', \ldots, \mathbf{K}_h') \quad (6)$$

$$\mathbf{V}' = (\mathbf{V}_1', \mathbf{V}_2', \ldots, \mathbf{V}_h') \quad (7)$$

$$\mathbf{I}' = (\mathbf{I}_1', \mathbf{I}_2', \ldots, \mathbf{I}_h'). \quad (8)$$

Next, we concatenate $\mathbf{K}_i'$ and $\mathbf{V}_i'$ with $\mathbf{I}_i'$ to form $\mathbf{K}$ and $\mathbf{V}$

$$\mathbf{K}_i = \text{Concat}(\mathbf{K}_i', \mathbf{I}_i'), \mathbf{V}_i = \text{Concat}(\mathbf{V}_i', \mathbf{I}_i'). \quad (9)$$

The $\mathbf{I}'$ retains local context and salient feature representations of $\mathbf{F}_2$, allowing the model to incorporate information from both $\mathbf{F}_1$ and $\mathbf{F}_2$ in a relatively light manner.

Subsequently, the ith attention score is computed using a scaled dot-product operation

$$\mathbf{Z}_i = \text{attention}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i) = \text{softmax}\left(\frac{\mathbf{Q}_i \mathbf{K}_i{}^T}{\sqrt{d}}\right)\mathbf{V}_i. \quad (10)$$

The final output of MHCA is obtained by concatenating all $\mathbf{Z}_i \in \mathbb{R}^{s\times(d/h)}$ and applying a projection as follows:

$$\mathbf{O} = \text{Concat}(\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_i, \ldots, \mathbf{Z}_n)\mathbf{W}^O \quad (11)$$

where $\mathbf{W}^O \in \mathbb{R}^{d\times d}$ and $\mathbf{O} \in \mathbb{R}^{s\times d}$.

In the CNN branch, we sequentially apply 3×3 and 5×5 convolutions. This hierarchical architecture first employs smaller window sizes in the initial layers to capture relevant features from local regions, and then it increases the window size to leverage information from longer-range dependencies.

After processing through both the CNN and transformer branches in parallel, we concatenate their outputs and apply a 1×1 convolutional layer to fuse local and non-local features. Notably, the LDCT reduces the number of feature channels sent to subsequent CNNs and transformers, thereby decreasing model complexity while allowing concurrent processing of local and non-local features for improved feature extraction.

D. Hybrid Loss Function

To assess topological structure information and enhance the similarity of positive pairs while reducing the similarity of negative pairs, we introduce a graph convolution contrast (GCC) strategy. This approach aids in aligning different views of samples within the feature distribution space. For effective training of our end-to-end framework, we develop a hybrid loss function that integrates supervised graph contrastive loss with classification loss.

1) Supervised Graph Contrastive Loss: Graph convolution (GC) is a widely used operator for processing unstructured data, directly operating on a graph to extract high-level features by aggregating neighborhood information from vertices. Formally, let $\mathbf{G} = (\hat{\mathbf{V}}, \mathbf{E}, \mathbf{A})$ represent an undirected graph, where $\hat{\mathbf{V}}$ and $\mathbf{E}$ denote the sets of vertices and edges, respectively, and $\mathbf{A}$ is the adjacency matrix. The weight of the edge between graph vertices i and j is defined using the radial basis function (RBF) as follows:

$$\mathbf{A}_{i,j} = \begin{cases} -e^{\alpha\|\mathbf{x}_i - \mathbf{x}_j\|^2}, & \mathbf{x}_j \in \mathbf{N}_i^K \\ 0, & \text{else} \end{cases} \quad (12)$$

where $\mathbf{x}_i$ and $\mathbf{x}_j$ are connected in the graph, α is a parameter controlling the spread of the RBF, and $\mathbf{N}_i^K$ is the set of K-nearest neighbors (KNNs) of $\mathbf{x}_i \in \hat{\mathbf{V}}$. This set consists of $\mathbf{x}_j \in \hat{\mathbf{V}}$ that have the smallest spectral-domain EDs, defined as $d(\mathbf{x}_i, \mathbf{x}_j) = \|\mathbf{x}_i - \mathbf{x}_j\|^2$. The adjacency matrix for updating the GC is given by

$$\mathbf{A} = \mathbf{D}^{-\frac{1}{2}}(\mathbf{A} + \mathbf{I})\mathbf{D}^{-\frac{1}{2}} \quad (13)$$

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where $\mathbf{D}$ is a diagonalized degree matrix with $\mathbf{D}_{i,i} = \sum_j \mathbf{A}_{i,j}$. $\mathbf{I}$ is a unit matrix with the same size as the graph vertices. The propagation rule for GC is expressed as follows:

$$f(\Theta_1, \Theta_2) = \sigma(\Theta_2 \xi(\sigma(\Theta_2 \mathbf{A}))\mathbf{A}) \tag{14}$$

where $\xi(\cdot)$ represents the dropout layer and $\sigma(\cdot)$ represents the ReLU activation function. Θ_1 and Θ_2 are trainable parameters of the GC.

We input the features extracted by the LDCT into the graph convolution network to calculate the graph contrastive loss. To generate different views of the samples, we apply a rotation operation to the data. Given that we know the label of each sample $\mathbf{z}_i^L$, we group samples with the same label in each batch. The average of these samples, denoted as $\mathbf{r}_l^p$, serves as the representation for the specific group, formulated as follows:

$$\mathbf{r}_l^p = \frac{\sum_{i=1}^{B} 1_{\{\hat{y}_i^p\}} \mathbf{z}_i^p}{T} \tag{15}$$

where 1 is an indicator function for samples with a pseudo-label equal to $l \in Y$ in each pathway $p \in \{1, 2\}$. T indicates the number of such samples in the batch, while B is the batch size.

To ensure class consistency between two groups sharing the same label across different pathways, we focus on generating similar representations for these groups in the feature space. Considering $(\mathbf{r}_i^1, \mathbf{r}_i^2)$ as positive pairs and $(\mathbf{r}_l^1, \mathbf{r}_m^p)$ as negative pairs with $m \in Y \backslash l$, the formulation for the supervised graph contrastive loss is as follows:

$$L_{\text{sgc}}(\mathbf{r}_l^1, \mathbf{r}_l^2) = -\log \frac{h(\mathbf{r}_l^1, \mathbf{r}_l^2)}{h(\mathbf{r}_l^1, \mathbf{r}_l^2) + \sum_{\substack{m=1 \\ p\in\{1,2\}}}^{\beta} 1_{\{m \neq l\}} h(\mathbf{r}_l^1, \mathbf{r}_m^p)} \tag{16}$$

where $h(u, v) = \exp((u^T v/\|u\|_2\|u\|_2)/\epsilon)$ represents the exponential of the cosine similarity, and ϵ is a temperature hyperparameter. β corresponds to the number of classes of the labeled samples.

2) Classification Loss: We utilize a classification loss to directly optimize the model for the classification task. Specifically, we minimize the cross-entropy loss (L_{cls}) between the predicted class probability $\hat{y}_s$ and the ground-truth label $\mathbf{y}_s$ as follows:

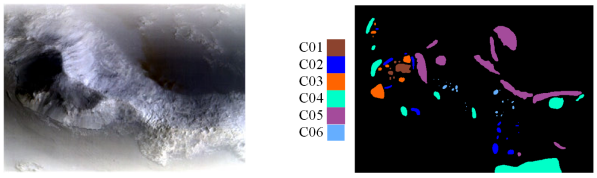
$$L_{\text{cls}} = -\sum_{s=1}^{S} \mathbf{y}_s \log \hat{y}_s \tag{17}$$

where S indicates the number of samples.

3) Loss Function: To enhance the discriminative features, we carefully design the overall loss function, which incorporates both the classification loss and the supervised graph contrastive loss as follows:

$$L = L_{\text{cls}} + \lambda L_{\text{sgc}} \tag{18}$$

where λ is the weight hyperparameter.

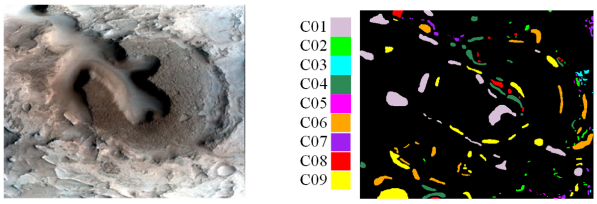


(a) (b)

Fig. 6. Holden Crater dataset. (a) False-color image. (b) Ground-truth map.

TABLE I
CATEGORY NAMES AND NUMBER OF SAMPLES DIVIDED IN THE HOLDEN CRATER DATASET

Class ID	Class Name	Train	Test	Total
1	Analcime	10	930	940
2	Plagioclase	10	1462	1472
3	Prehnite	10	1550	1560
4	High-Ca Pyroxene	10	6953	6963
5	Serpentine	10	8687	8697
6	Margarite	10	448	458
	Total	60	20030	20090



(a) (b)

Fig. 7. Nili Fossae dataset. (a) False-color image. (b) Ground-truth map.

TABLE II
CATEGORY NAMES AND NUMBER OF SAMPLES DIVIDED IN THE NILI FOSSAE DATASET

Class ID	Class Name	Train	Test	Total
1	Fe-Olivine	10	8553	8563
2	Epidote	10	986	996
3	Chlorite	10	376	386
4	Bassanite	10	3276	3286
5	Illite/Muscovite	10	218	228
6	Mg-Carbonate	10	4857	4867
7	Plagioclase	10	1047	1057
8	Prehnite	10	1151	1161
9	Serpentine	10	6156	6166
	Total	90	26620	26710

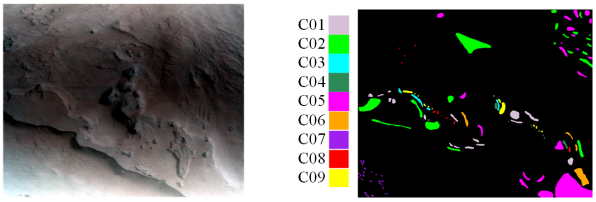
IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Datasets and Evaluation Measures

To assess the effectiveness of our proposed method, we compiled a collection of Martian HSIs and annotated them ourselves.[1] We then performed comprehensive experiments on these datasets, which include scenes from Holden Crater (HC), Nili Fossae (NF), and Utopia Planitia (UP). The data was obtained from the compact reconnaissance imaging spectrometer for Mars (CRISM) aboard the Mars reconnaissance orbiter (MRO). The CRISM data boasts a high spatial resolution of 18 m/pixel and includes 544 spectral bands ranging from 0.36 to 3.93 μm, making it particularly suitable for mineral detection on the Martian surface [49].

[1] Available at: https://doi.org/10.57760/sciencedb.19732.

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(a) (b)

Fig. 8. Utopia Planitia dataset. (a) False-color image. (b) Ground-truth map.

TABLE III
CATEGORY NAMES AND NUMBER OF SAMPLES DIVIDED IN THE UTOPIA PLANITIA DATASET

Class ID	Class Name	Train	Test	Total
1	Analcime	10	1741	1751
2	Bassanite	10	5797	5807
3	High-Ca Pyroxene	10	422	432
4	Illite/Muscovite	10	262	272
5	Low-Ca Pyroxene	10	6764	6774
6	Mg-Smectite	10	1156	1166
7	Monohydrated sulfate	10	330	340
8	Plagioclase	10	192	202
9	Prehnite	10	584	594
	Total	90	17248	17338

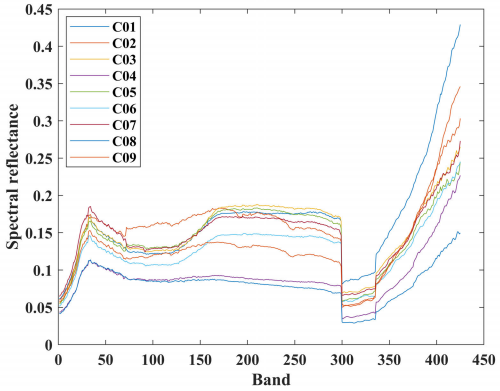


Fig. 9. Spectral curves of different mineral classes from the NF dataset.

For the three datasets, the false-color map and ground-truth are displayed in Figs. 6–8, respectively. Different colors in each dataset represent different minerals. Class names and detailed sample numbers are presented in Tables I–III, respectively. Specifically, the HC is notable for its fluviolacustrine geological evolution, although the specifics of this evolution, such as water level changes and timing, remain unclear. The HC dataset consists of 418 × 595 pixels and retains 440 bands after filtering out water absorption and noise spectra. NF is unique for containing additional mineral phases such as carbonates and serpentine, which suggests possible hydrothermal activity—an important factor for the potential emergence of biological activity on Mars. The NF dataset has dimensions of 478 × 593 pixels and includes 425 bands after discarding several noisy channels [50]. UP, the landing site for the Tianwen-1 rover "Zhurong" [51], has a dataset size of 478 × 595 × 432.

Exploring Mars presents greater challenges compared to Earth observation. The limited availability of labeled samples due to insufficient prior knowledge complicates the process.

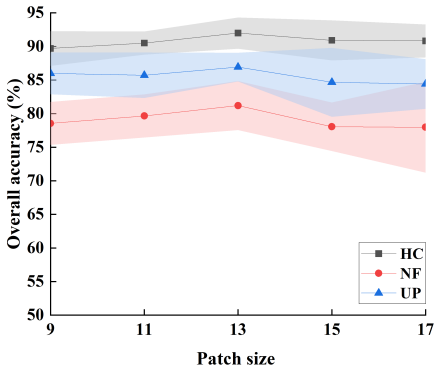


Fig. 10. OA with standard deviation on the HC, NF, and UP for different patch sizes.

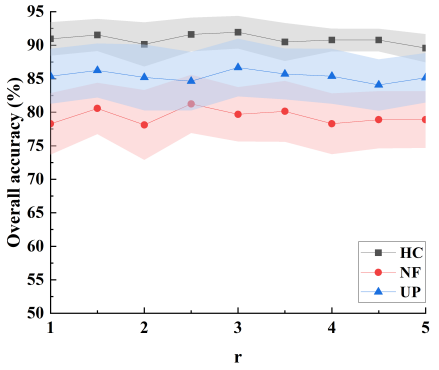


Fig. 11. OA with standard deviation on the HC, NF, and UP for different r.

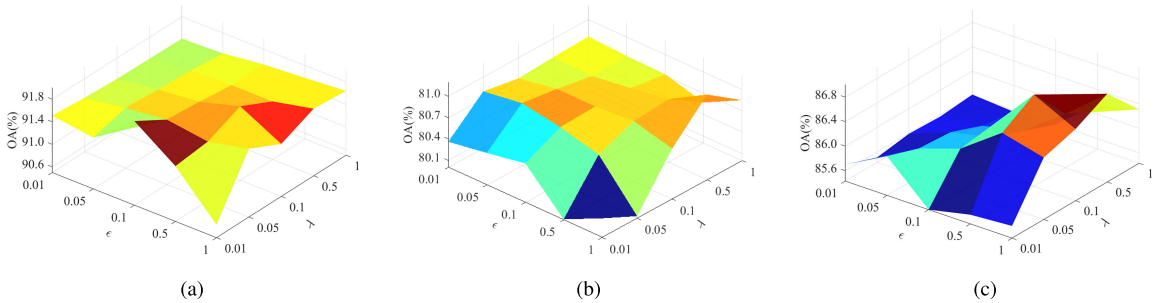
Additionally, Fig. 9 displays the spectral curves of different mineral classes from the NF dataset. It is observed that different minerals have high inter-class similarity and high dimensions, which will hinder accurate classification results. To address these challenges, we developed a new algorithm, MCTGCL, specifically for HyMars classification, aimed at improving the feature extraction capabilities of the networks. For performance evaluation, we adopt the class-specified accuracy (CA), average accuracy (AA), overall accuracy (OA), and Kappa coefficient (Kappa) as metrics.

B. Experimental Settings

To maximize the effectiveness of the proposed networks, we implemented detailed experimental configurations. We set the total number of training episodes to 100 and established the learning rate at 0.001, adhering to standard practices. Furthermore, to evaluate the effect of patch size on classification performance, we kept other parameters constant and tested a range of patch sizes: 9, 11, 13, 15, and 17. Fig. 10 depicts the OA of the model across these different patch sizes for the three datasets, with a patch size of 13 yielding the best results.

Additionally, we conducted experiments to identify the ratio $r = (c/c_p)$, a hyperparameter that regulates the channel dimension used in convolution within PConv, which affects the number of parameters. The experimental findings presented in Fig. 11 reveal that the OAs for the HC and UP datasets achieve optimal performance when r is set to 3, while the NF dataset performs best with r at 2.5.

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(a) (b) (c)

Fig. 12. Parameter tuning of λ and ϵ for the proposed network in the three datasets. (a) HC. (b) NF. (c) UP.

TABLE IV
CLASSIFICATION PERFORMANCE (%) OF THE ABLATIONS ON THE PROPOSED METHOD ON HC, NF, AND UP

Data set	Case	IEAM	LDCT	L_{sgc}	AA	OA	Kappa
HC	1				91.86 ± 1.37	90.00 ± 2.42	85.58 ± 3.36
	2	✓			92.40 ± 1.91	90.78 ± 3.32	86.71 ± 4.67
	3	✓	✓		92.64 ± 1.45	91.32 ± 2.84	87.43 ± 4.02
	4	✓	✓	✓	**93.30 ± 1.40**	**91.99 ± 2.32**	**88.39 ± 3.26**
NF	1				80.64 ± 3.00	78.30 ± 4.31	72.97 ± 5.20
	2	✓			80.81 ± 3.55	79.44 ± 4.36	74.47 ± 5.24
	3	✓	✓		82.43 ± 3.16	80.80 ± 4.00	76.09 ± 4.85
	4	✓	✓	✓	**82.68 ± 3.09**	**81.20 ± 3.65**	**76.57 ± 4.45**
UP	1				87.12 ± 3.31	85.06 ± 4.75	79.83 ± 6.05
	2	✓			88.07 ± 2.28	85.71 ± 3.48	80.61 ± 4.45
	3	✓	✓		88.24 ± 1.69	86.15 ± 3.31	81.16 ± 4.22
	4	✓	✓	✓	**88.68 ± 1.55**	**86.93 ± 2.14**	**82.19 ± 2.76**

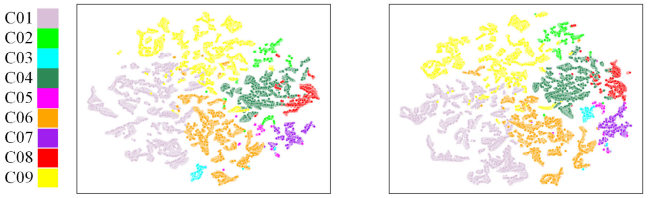
Moreover, we conducted extensive experiments to evaluate the influence of hyperparameters related to the supervised graph contrastive loss across the three test datasets. Specifically, we examined the weight hyperparameter λ and the temperature factor ϵ. We selected values for λ and ϵ from the set $\{0.01, 0.05, 0.1, 0.5, 1\}$, and investigated their mutual effects on performance. As depicted in Fig. 12, the optimal results were achieved with the pairs λ and ϵ set to (0.01, 0.1), (0.5, 1), and (0.1, 0.5) for the HC, NF, and UP datasets, respectively. These experiments demonstrate that the outcomes are significantly affected by these two hyperparameters, particularly the weight factor λ, which is crucial in determining the influence of the supervised graph contrastive loss.

C. Ablations on the Method

As outlined in Section III, the proposed approach primarily consists of the IEAM and LDCT for feature extraction, along with a hybrid loss function that integrates the classification loss and supervised graph contrastive loss. To assess the effectiveness of each component, we performed ablation studies focusing on these specific modules.

We define a baseline network that excludes IEAM, LDCT, and the L_{sgc}. This baseline comprises a standard channel attention module akin to SE and a dual-branch CNN–transformer that processes the entire input data in parallel. We then incrementally incorporate IEAM and LDCT by substituting the existing modules in the baseline with these components. Finally, we integrate L_{sgc} to form the complete architecture. The four experimental cases are as follows.

Case 1: The proposed network without IEAM, LDCT, and L_{sgc}.



(a) (b)

Fig. 13. Visualization of embedded feature vectors by t-SNE maps on the NF dataset in different cases. (a) Case 3. (b) Case 4.

Case 2: The proposed network without LDCT and L_{sgc}.

Case 3: The proposed network without L_{sgc}.

Case 4: The proposed MCTGCL that includes IEAM, LDCT, and L_{sgc}.

For each class, we randomly selected 10 labeled samples, and the experiments were conducted independently ten times. The average classification performance along with the standard deviation for the three datasets is summarized in Table IV. Notably, the best and second-best results for OA, AA, and Kappa are highlighted in bold and underlined, respectively. From Table IV, it is evident that our proposed IEAM, LDCT, and L_{sgc} can significantly contribute to the improvement of OA, AA, and Kappa. Specifically, IEAM incorporates direction-specific information into the channel attention along the spatial dimension, leading to a more comprehensive aggregation of features. Meanwhile, LDCT effectively captures both local and global features for HSIC with relatively low complexity. Additionally, the hybrid loss function proves to be essential for optimizing the feature space by leveraging the graph structure of labeled samples and identifying similarities across different views. Furthermore, we visually inspect the feature vectors extracted in cases 3 and 4 using the t-SNE method. Fig. 13 illustrates the feature distributions for the NF dataset. It can be observed that the features from the same class are more tightly clustered, particularly for class 2 and class 7 samples. This observation underscores the significant role of supervised graph contrastive loss in enhancing the extraction of discriminative features.

D. Comparison With Other Methods

To validate the effectiveness of the proposed MCTGCL, we conduct a comparative analysis with the well-established

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TABLE V
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE HC DATASET

No.	3D-CNN [20]	SSRN [21]	A^2S^2K-ResNet [24]	SpectralFormer [26]	GAHT [27]	SSFTT [43]	MorphFormer [45]	MCTGCL
1	98.05 ± 2.03	96.87 ± 5.14	97.62 ± 3.64	96.42 ± 4.87	97.50 ± 3.08	98.48 ± 2.42	**99.53 ± 0.59**	99.00 ± 0.93
2	76.68 ± 7.15	86.51 ± 6.84	84.27 ± 8.42	83.68 ± 7.44	82.93 ± 5.46	87.98 ± 6.17	82.11 ± 8.33	**90.68 ± 5.74**
3	78.69 ± 5.92	**92.57 ± 5.24**	87.96 ± 5.14	81.39 ± 7.24	86.40 ± 5.84	86.15 ± 7.10	85.02 ± 5.66	88.43 ± 8.38
4	73.91 ± 8.27	86.08 ± 7.12	83.11 ± 7.64	74.58 ± 5.06	79.49 ± 7.76	82.81 ± 5.32	75.71 ± 10.66	**87.49 ± 5.77**
5	83.62 ± 7.16	87.18 ± 6.39	72.59 ± 8.98	86.10 ± 5.74	92.73 ± 5.62	86.93 ± 7.73	82.56 ± 5.68	**95.33 ± 3.59**
6	**99.49 ± 0.44**	99.33 ± 0.88	95.89 ± 3.82	99.31 ± 0.69	98.77 ± 1.55	99.04 ± 0.86	97.88 ± 0.94	98.88 ± 1.02
AA	85.08 ± 3.18	91.42 ± 2.48	86.91 ± 2.29	86.91 ± 1.09	89.63 ± 1.67	90.23 ± 1.82	87.14 ± 2.56	**93.30 ± 1.40**
OA	80.39 ± 5.86	87.89 ± 4.23	79.97 ± 2.99	82.33 ± 2.26	87.28 ± 1.97	86.32 ± 3.51	81.47 ± 4.46	**91.99 ± 2.32**
Kappa	72.53 ± 7.88	82.76 ± 5.86	72.11 ± 3.80	75.14 ± 3.03	81.81 ± 2.69	80.50 ± 4.86	73.73 ± 5.80	**88.39 ± 3.26**

TABLE VI
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE NF DATASET

No.	3D-CNN [20]	SSRN [21]	A^2S^2K-ResNet [24]	SpectralFormer [26]	GAHT [27]	SSFTT [43]	MorphFormer [45]	MCTGCL
1	64.40 ± 11.01	74.04 ± 10.98	74.21 ± 9.65	83.32 ± 8.20	75.97 ± 11.61	73.08 ± 9.22	60.77 ± 8.31	**87.06 ± 6.96**
2	82.95 ± 5.76	82.20 ± 7.62	83.63 ± 6.55	85.16 ± 4.72	79.34 ± 6.62	80.59 ± 6.35	75.84 ± 8.63	**88.78 ± 6.56**
3	**92.26 ± 3.13**	85.93 ± 6.17	87.34 ± 6.16	91.49 ± 4.53	79.55 ± 17.41	91.46 ± 4.23	88.59 ± 4.94	92.07 ± 4.10
4	63.61 ± 10.37	76.09 ± 9.82	73.44 ± 10.75	61.68 ± 10.21	68.19 ± 11.40	66.83 ± 7.49	70.73 ± 8.63	**76.96 ± 8.23**
5	66.05 ± 9.68	77.11 ± 15.17	79.63 ± 11.35	70.73 ± 10.13	76.24 ± 15.29	71.06 ± 12.60	72.06 ± 12.74	**80.60 ± 10.57**
6	52.49 ± 6.98	59.99 ± 10.49	53.46 ± 6.18	51.87 ± 9.79	63.00 ± 8.87	60.64 ± 8.69	51.13 ± 7.24	**66.38 ± 8.46**
7	67.27 ± 10.83	79.88 ± 7.81	79.13 ± 8.70	73.47 ± 10.14	80.08 ± 10.77	**84.33 ± 6.97**	70.18 ± 12.23	83.99 ± 9.16
8	80.20 ± 8.39	**85.78 ± 7.25**	84.90 ± 7.51	83.64 ± 7.18	83.89 ± 8.62	80.65 ± 6.40	80.05 ± 7.63	84.17 ± 8.50
9	58.94 ± 12.07	75.02 ± 12.85	72.66 ± 10.97	72.88 ± 11.64	79.98 ± 10.32	75.55 ± 10.43	63.47 ± 13.83	**84.11 ± 9.56**
AA	69.80 ± 2.90	77.34 ± 2.48	76.49 ± 1.98	74.92 ± 2.46	76.25 ± 5.44	76.02 ± 3.50	70.31 ± 2.40	**82.68 ± 3.09**
OA	62.76 ± 5.20	73.19 ± 3.96	71.20 ± 3.60	72.21 ± 4.29	74.25 ± 5.75	71.91 ± 4.45	63.11 ± 5.01	**81.20 ± 3.65**
Kappa	54.77 ± 5.88	67.02 ± 4.66	64.72 ± 4.20	65.79 ± 5.04	68.42 ± 6.79	65.44 ± 5.19	55.41 ± 5.48	**76.57 ± 4.45**

TABLE VII
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE UP DATASET

No.	3D-CNN [20]	SSRN [21]	A^2S^2K-ResNet [24]	SpectralFormer [26]	GAHT [27]	SSFTT [43]	MorphFormer [45]	MCTGCL
1	59.23 ± 7.86	**81.83 ± 7.81**	74.05 ± 8.80	59.06 ± 8.16	73.02 ± 6.09	73.95 ± 8.24	75.82 ± 5.49	75.95 ± 8.54
2	72.24 ± 7.93	75.57 ± 11.12	75.79 ± 11.77	73.42 ± 12.07	77.39 ± 14.27	79.95 ± 9.20	76.04 ± 7.72	**82.73 ± 6.07**
3	84.62 ± 7.64	82.75 ± 13.06	**92.44 ± 7.58**	89.57 ± 10.05	85.59 ± 5.58	91.59 ± 7.39	89.01 ± 6.90	90.76 ± 8.00
4	89.12 ± 8.61	90.54 ± 5.99	93.24 ± 6.74	91.68 ± 5.83	91.79 ± 7.29	**93.70 ± 5.00**	90.30 ± 3.76	93.63 ± 6.53
5	84.07 ± 5.55	81.05 ± 6.34	85.48 ± 5.12	84.87 ± 6.19	85.95 ± 7.47	86.00 ± 5.20	83.31 ± 5.84	**91.26 ± 5.11**
6	90.26 ± 5.93	94.07 ± 6.66	93.09 ± 7.09	93.37 ± 5.53	90.68 ± 8.42	92.86 ± 6.77	92.86 ± 7.22	**94.39 ± 6.65**
7	98.64 ± 3.33	**100.00 ± 0.00**	99.91 ± 0.27	**100.00 ± 0.00**	99.94 ± 0.18	**100.00 ± 0.00**	99.70 ± 0.81	**100.00 ± 0.00**
8	76.72 ± 13.61	84.38 ± 15.51	**90.05 ± 11.52**	84.01 ± 15.45	86.87 ± 10.46	81.25 ± 18.16	80.10 ± 16.01	85.73 ± 14.80
9	76.75 ± 7.46	**85.24 ± 8.60**	81.41 ± 7.88	84.52 ± 7.18	78.32 ± 9.12	82.40 ± 4.93	80.07 ± 6.90	83.68 ± 5.21
AA	81.29 ± 2.57	86.16 ± 1.54	87.27 ± 1.36	84.50 ± 1.90	85.51 ± 2.89	86.86 ± 2.21	85.25 ± 1.51	**88.68 ± 1.55**
OA	78.04 ± 3.36	80.89 ± 3.32	82.06 ± 4.19	79.47 ± 3.90	82.18 ± 4.91	83.56 ± 3.59	81.16 ± 2.26	**86.93 ± 2.14**
Kappa	70.85 ± 4.08	74.44 ± 4.11	75.97 ± 5.27	72.56 ± 4.75	76.01 ± 6.20	77.85 ± 4.53	74.63 ± 2.77	**82.19 ± 2.76**

CNN-based spectral–spatial classifiers, such as 3D-CNN [20] and SSRN [21]. The A^2S^2K-ResNet [24] that utilizes CNN-based attention mechanism is also employed for comparison. Furthermore, we analyze the performance of transformer-based models such as SpectralFormer [26], GAHT [27], SSFTT [43], and MorphFormer [45], which focus on global information extraction. For the training process, ten samples are randomly selected from each class. To ensure the reliability of our experimental findings, each method and dataset undergoes ten independent runs, with the mean accuracies and standard deviations reported in Tables V–VII. The highest and second-best accuracies in each row are highlighted in bold and underlined, respectively, facilitating easier comparison and interpretation of the results.

Table V illustrates the classification performance of various methods on the HC dataset. Notably, MCTGCL surpasses other methods in terms of AA, OA, and Kappa metrics. It achieves the top position in three categories and ranks second in two, with improvements over the second-best values of 1.88%, 4.10%, and 5.63% for AA, OA, and Kappa, respectively. This indicates that our method is adept at extracting more discriminative information from the original data, enhancing the ability to differentiate between various data features and yielding promising results. Moving on to Table VI, which provides the classification results on the NF dataset, it can be observed that the proposed MCTGCL demonstrates superior classification performance compared to other methods. Particularly, MCTGCL secures the first and second positions in six and two categories, respectively, showcasing its effectiveness in this dataset. Furthermore, Table VII summarizes the classification accuracies for the UP dataset. The MCTGCL achieves the top-level classification performance, again demonstrating the capacity of the proposed framework. Noteworthy is that the AA, OA, and Kappa metrics

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TABLE VIII
PARAMETERS AND TRAINING TIME OF THE PROPOSED METHOD AND THE COMPETITIVE DEEP LEARNING-BASED METHODS

		3DCNN	SSRN	A^2S^2K-ResNet	SpectralFormer	GAHT	SSFTT	MorphFormer	MCTGCL
Params		132.7K	135.9K	105.3K	130.1K	946.8K	152.8K	141.7K	247.0K
Training time (s)	HC	1.55	2.20	2.72	21.72	3.38	1.82	4.19	64.53
	NF	2.90	4.47	4.72	28.83	6.08	3.47	7.61	136.44
	UP	2.69	4.11	4.62	29.70	6.41	3.31	7.82	92.11

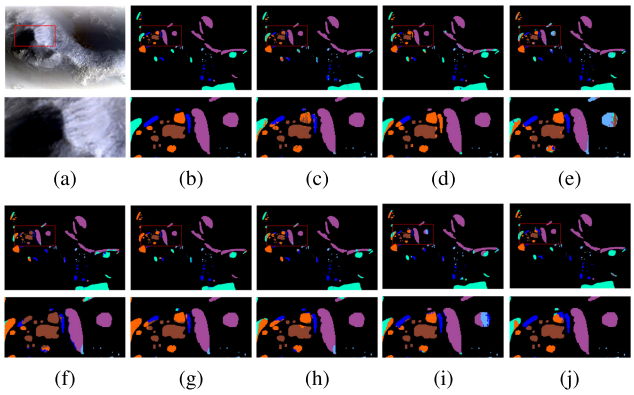


Fig. 14. Classification maps obtained via various algorithms for the HC dataset. (a) False-color image. (b) Ground-truth. (c) 3D-CNN. (d) SSRN. (e) A^2S^2K-ResNet. (f) SpectralFormer. (g) GAHT. (h) SSFTT. (i) MorphFormer. (j) MCTGCL.

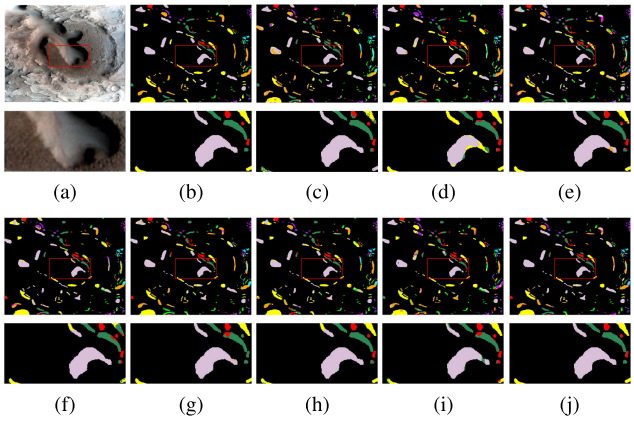


Fig. 15. Classification maps obtained via various algorithms for the NF dataset. (a) False-color image. (b) Ground-truth. (c) 3D-CNN. (d) SSRN. (e) A^2S^2K-ResNet. (f) SpectralFormer. (g) GAHT. (h) SSFTT. (i) MorphFormer. (j) MCTGCL.

exceed the second-best values by 1.41%, 3.37%, and 4.34%, respectively, further highlighting the effectiveness of the MCTGCL frameworks.

Overall, both CNN-based and transformer-based methods encounter performance challenges in HSIC tasks related to Mars. Therefore, it is essential to consider the extraction of both local and global information while enhancing the aggregation of useful data. In particular, the MCTGCL consistently achieves the highest performance across all three datasets, underscoring its efficacy in HSIC tasks.

To facilitate visual comparison, we present the optimal classification maps generated by various methods across the three datasets from Figs. 14–16, alongside the data presented

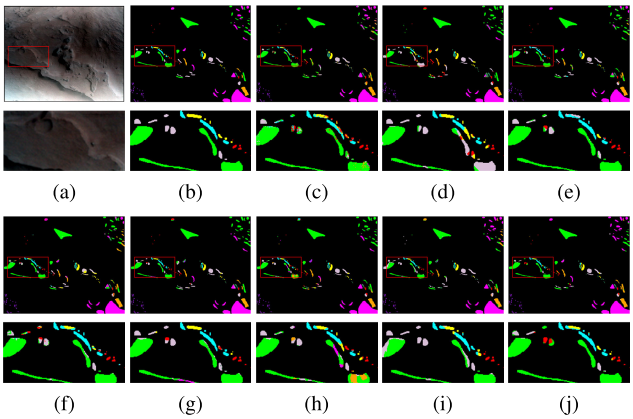


Fig. 16. Classification maps obtained via various algorithms for the UP dataset. (a) False-color image. (b) Ground-truth. (c) 3D-CNN. (d) SSRN. (e) A^2S^2K-ResNet. (f) SpectralFormer. (g) GAHT. (h) SSFTT. (i) MorphFormer. (j) MCTGCL.

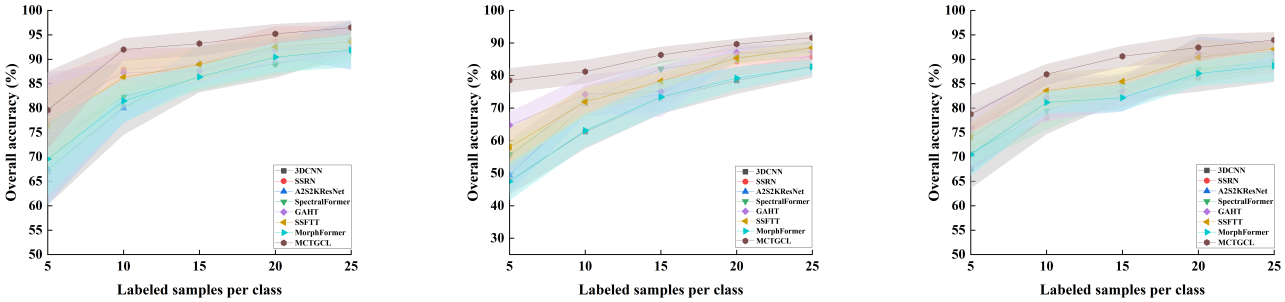
in Tables V–VII. These figures include the corresponding composite false-color images and ground-truth for a comprehensive evaluation. In Fig. 14, it is clear that the classification map produced by our method effectively distinguishes between different mineral categories in small areas with multiple categories. For instance, the area in the top left corner of the map shows a relatively satisfactory result with MCTGCL. Similarly, in Fig. 15 for the NF dataset, the classification map generated by MCTGCL closely aligns with the ground-truth and exhibits minimal noise. This phenomenon demonstrates the ability of the approach to extract distinctive inter-class features—a trend that is further confirmed in Fig. 16.

Furthermore, to assess the effectiveness of the proposed method with varying labeled samples, we randomly select annotated samples in quantities of 5, 10, 15, 20, and 25 and conduct the experiments ten times. The mean OAs and standard deviations of various methods on the three datasets are illustrated in Fig. 17. Remarkably, the MCTGCL ranks first place in all scenarios, affirming the robustness of the proposed network and its capability to extract effective and discriminative features.

E. Analysis of the Computational Complexity

This section explores the computational complexity of the MCTGCL. Specifically, we conduct a comparative analysis focusing on the number of parameters (para.) and the training time of the MCTGCL and the other models. All experiments are performed on the NVIDIA GeForce GTX 4090 GPU. The parameter statistics and training durations are summarized in Table VIII. From Table VIII, it can be observed that the

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(a) (b) (c)

Fig. 17. OA (%) of various methods on the three datasets with different numbers of labeled samples. (a) HC. (b) NF. (c) UP.

MCTGCL has a moderate number of parameters compared to alternative methods, allowing it to extract more effective features while reducing the risk of overfitting. Notably, the MCTGCL exhibits the longest training time across the three datasets, indicating that the network requires more time for training. We attribute this extended training duration to the computation of the supervised graph contrastive L_{sgc}. Specifically, the training time for the HC dataset is significantly shorter than for the NF and UP datasets, likely due to the fewer number of classes present in HC.

V. Conclusion

This article proposes an MCTGCL to boost the classification performance. Initially, a shallow CNN module and an IEAM are proposed to extract shallow spatial–spectral features and incorporate spatial information into channel attention, respectively. Following this, we introduce the LDCT network, which effectively captures both local and global features. Additionally, a supervised graph contrastive loss function is developed, which constructs the graph structure of labeled samples from two different views, encouraging the network to learn more discriminative features in the latent space. To validate the effectiveness of our proposed method, we create three Mars HSI datasets (HyMars) and focus on their classification. Extensive experimental results demonstrate the superiority of the proposed MCTGCL in handling Mars HSIC tasks. Our future work will focus on developing more lightweight models that are more efficient and maintain the classification performance.

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Bobo Xi (Member, IEEE) received the B.E. degree in information engineering and the Ph.D. degree in information and communication engineering from Xidian University, Xi'an, China, in 2017 and 2022, respectively.

He is currently a Lecturer with the State Key Laboratory of Integrated Services Networks, Xidian University. He has published more than 30 articles in refereed journals, including IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His research interests include hyperspectral image processing, machine learning, and deep learning.



Yun Zhang (Graduate Student Member, IEEE) received the B.E. degree in communication engineering from Xidian University, Xi'an, China, in 2024, where she is currently pursuing the M.E. degree in information and communication engineering.

Her research interests include hyperspectral image processing and deep learning.



Jiaojiao Li (Senior Member, IEEE) received the B.E. degree in computer science and technology, the M.S. degree in software engineering, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2009, 2012, and 2016, respectively.

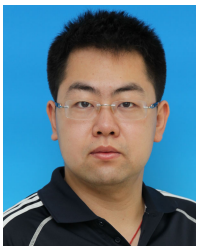
She was an Exchange Ph.D. Student with Mississippi State University, Starkville, MS, USA. She is currently a Professor and a Ph.D. Supervisor with Xidian University. Her research interests include hyperspectral remote sensing image analysis and processing, pattern recognition, and data compression.

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Tie Zheng received the bachelor's degree from Northeast Agricultural University, Harbin, China, in 2017, and the Ph.D. degree from the University of the Chinese Academy of Sciences, Beijing, China, in 2023.

He is currently employed with the National Space Science Center, Chinese Academy of Sciences, Beijing, with major research on space signal processing and remote sensing techniques.



Xunfeng Zhao received the B.E. degree in automation and the M.S. degree in automatic control from Northwestern Polytechnical University, Xi'an, China, in 2004 and 2007, respectively. He is currently pursuing the Ph.D. degree in electronic information with Xidian University, Xi'an.

He has served as the Chief Designer of the scientific payload subsystem for the "Queqiao-2" relay satellite and the Technical Leader of the effective payload subsystem for the "Tianwen-3" planetary exploration mission. His research interests include the design of aerospace scientific payload systems and aerospace vehicle software design technology.



Haitao Xu received the B.E. degree in electrical engineering and automation and the M.S. degree in power electronics and power drives from Beijing Jiaotong University, Beijing, China, in 2005 and 2008, respectively. He is currently pursuing the Ph.D. degree in computer application technology with the University of Chinese Academy of Sciences, Beijing.

He has served as the Deputy Chief Designer of the payload subsystem for the Chang'e-4 lunar exploration mission and the payload subsystem for the Tianwen-2 planetary exploration mission, and the Chief Designer of the payload subsystem for the Chang'e-8 lunar exploration mission. His research interests include aerospace-integrated electronic system technology and space intelligent information processing technology.



Changbin Xue received the M.S. degree from Harbin Institute of Technology, Harbin, China, in 1997, and the Ph.D. degree from Beijing Institute of Technology, Beijing, China, in 2017.

He has been engaged in the design and development of complex space electronic information systems and space science exploration payload systems for more than 20 years. He served as the Chief Designer of the Chang'e-4 payload system and the space science pilot project "Shijian-10 Return Science Experiment Satellite." He has published more than 30 academic papers and more than ten authorized patents. His research interests include full process design and simulation technology of deep space exploration scientific payload, machine learning, and deep learning.



Yunsong Li (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

In 1999, he joined the School of Telecommunications Engineering, Xidian University, where he is currently a Professor. He is also the Director of the State Key Laboratory of Integrated Service Networks, Image Coding and Processing Center. His research interests include image and video processing, hyperspectral image processing, and high-performance computing.



Jocelyn Chanussot (Fellow, IEEE) received the M.Sc. degree in electrical engineering from Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Universite de Savoie, Annecy, France, in 1998.

From 1999 to 2023, he was with Grenoble INP, where he was a Professor of signal and image processing. From 2015 to 2017, he was a Visiting Professor with the University of California at Los Angeles (UCLA), Los Angeles, CA, USA. He is currently a Research Director with INRIA, Grenoble. He has been a Visiting Scholar with Stanford University, Stanford, CA, USA; KTH, Stockholm, Sweden; and NUS, Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavík, Iceland. He holds the AXA Chair in remote sensing and is an Adjunct Professor with the Aerospace Information research Institute, Chinese Academy of Sciences, Beijing. His research interests include image analysis, hyperspectral remote sensing, data fusion, machine learning, and artificial intelligence.

Dr. Chanussot is a fellow of an ELLIS Fellow and the Asia-Pacific Artificial Intelligence Association. He was a member of the Machine Learning for Signal Processing Technical Committee of the IEEE Signal Processing Society from 2006 to 2008. He was a member of the Institut Universitaire de France from 2012 to 2017, and has been a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters) since 2018. He was the Founding President of IEEE Geoscience and Remote Sensing French Chapter from 2007 to 2010, which received the 2010 IEEE GRSS Chapter Excellence Award. He has received multiple outstanding paper awards. He was the Vice-President of the IEEE Geoscience and Remote Sensing Society, in charge of meetings and symposia, from 2017 to 2019. He was the General Chair of the first IEEE GRSS Workshop on Hyperspectral Image and Signal Processing, Evolution in Remote sensing (WHISPERS). From 2009 to 2011, he was the Chair and the Co-Chair of the GRSS Data Fusion Technical Committee from 2005 to 2008. He has been the Program Chair of the IEEE International Workshop on Machine Learning for Signal Processing since 2009. He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON IMAGE PROCESSING, and PROCEEDINGS OF THE IEEE. He was the Editor-in-Chief of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2011 to 2015. In 2014, he served as a Guest Editor for *IEEE Signal Processing Magazine*.

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